

Investigating Postoperative Hypoxemia Using the Medical Informatics Operating Room Vitals and Events Repository (MOVER)

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Abstract

- MOVER data were wrangled to find significant predictors of postoperative hypoxemia using a full logistic regression model
 - Male sex and higher body mass index (BMI) were identified
- A reduced logistic regression model was then fitted solely using sex and BMI as predictors
 - The test set misclassification error rate for the reduced model was 28.0%, slightly lower than that for the full model
- The results suggest that high-BMI or male patients should be paid greater attention to improve clinical outcomes
- Also, including more variables when modelling may not be clinically beneficial

Introduction & Literature Review

- Postoperative hypoxemia is a frequent complication from anesthetic¹
- The MOVER EPIC dataset includes records from 64,354 surgeries between 2017 and 2022² and can be used to study this issue
- Prior research has assessed how postoperative hypoxemia is affected by different variables, but these models only contained up to four laboratory test results^{3,4,5}
- To fill this gap characterized by few laboratory test results, this work modelled postoperative hypoxemia probability using logistic regression with up to ten preoperative laboratory test results and three demographic variables as covariates
- The objective was to detect significant predictors of postoperative hypoxemia and determine the accuracy of predicting its probability with just these variables

Methodology (The Pipeline)

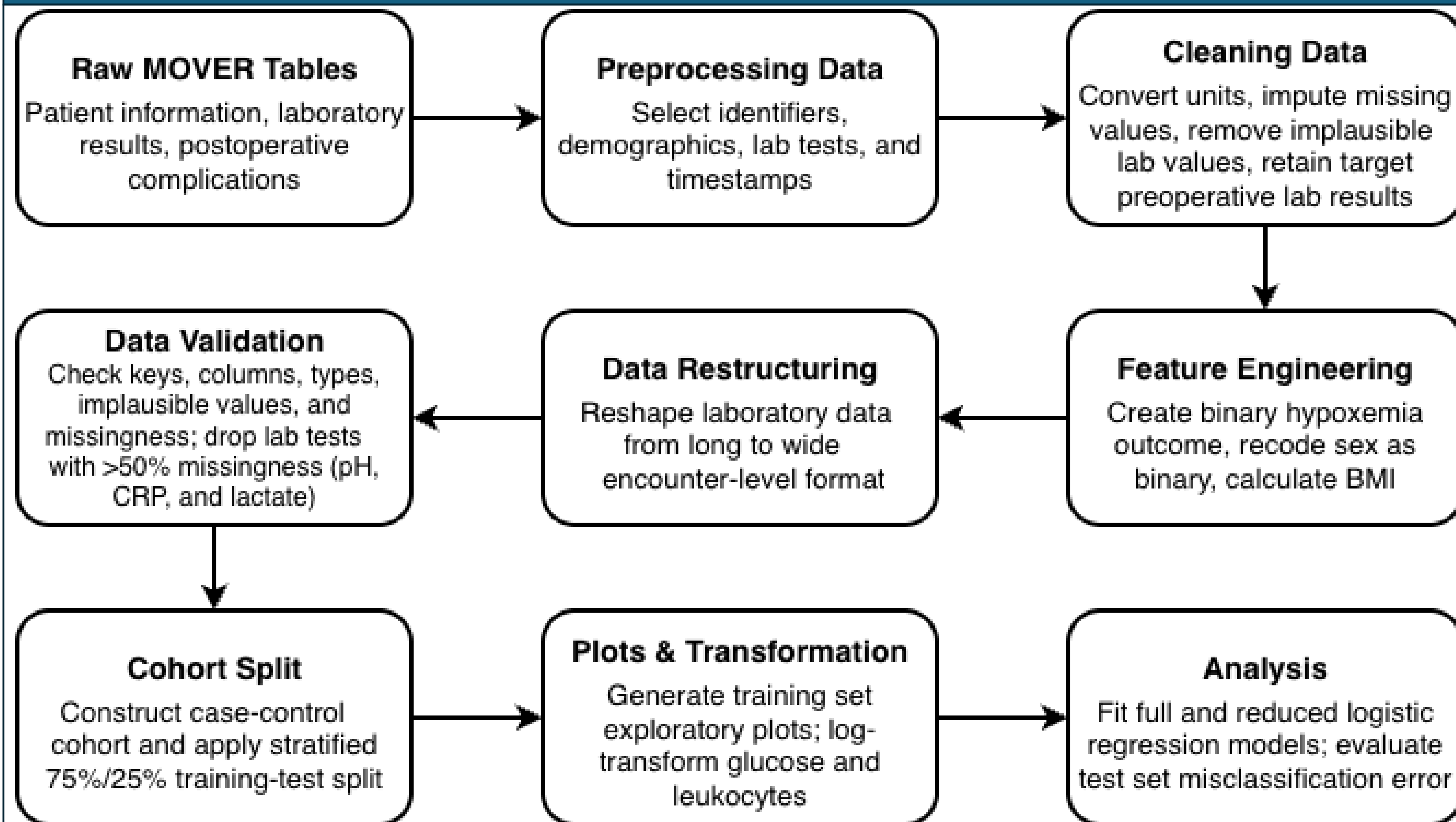


Figure 1. Flow chart of data wrangling architecture [CRP = C-reactive protein]

Key Findings

Table 1. Full logistic regression model results

Variable	Coefficient Estimate	P-value
Intercept	-6.576	0.237
Age	0.004	0.660
Sex: Female	-1.426	0.001
Body mass index	0.051	0.040
Carbon dioxide	0.015	0.716
log(Glucose)	0.786	0.114
Hematocrit	0.094	0.557
Hemoglobin	-0.521	0.264
log(Leukocytes)	0.257	0.485
Potassium	0.117	0.717
Sodium	0.014	0.650

- In the full model, sex and body mass index (BMI) were the only predictors significant at the 0.05 level (Table 1)
- Female sex and higher BMI were associated with lower and higher odds of postoperative hypoxemia, respectively
- The reduced model, which included only sex and BMI as predictors, achieved a lower test set misclassification error rate (28.0%) than the full model (29.3%)

Conclusion & Impact

Key Takeaway: Patients who are **male** or **have high BMI** may benefit from closer postoperative monitoring for timely detection and intervention of hypoxemia

Key Risk Factors for Postoperative Hypoxemia

- **Higher BMI** elevates odds, aligning with established literature⁴
- Higher odds among **males**, contradicting previous findings⁴

Model Performance and Limitation

- The reduced model outperformed the full model, suggesting that additional variables introduced noise rather than clinical signal
- Only 98 out of 24,167 entries (0.4%) had hypoxemia; this extreme class imbalance limited model performance despite undersampling control cases to balance the data

Future Directions

- Employing systematic oversampling or re-weighting techniques to improve minority class classification
- Investigating alternative biomarkers and the interplay between hypoxemia and concurrent postoperative complications to enhance predictive accuracy

References

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